Mapping the Social, Behavioral, and Psychosocial Exposome and Incident Mental and Metabolic Health Using Causal Discovery

Kate T. Tran, MPH[1,2]; Vi H. Tran, DAOM[3]; Ran Barzilay, MD PhD[1,2,4]; Fares Alahdab MD, MS, MSc, FAHA, FACC[5,6]

[1]Department of Child and Adolescent Psychiatry and Behavioral Sciences, Children's Hospital of Philadelphia (CHOP); [2]Lifespan Brain Institute of CHOP and Penn Medicine; [3]VNVC, Ho Chi Minh City, Vietnam; [4]Department of Psychiatry, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA; [5]Department of Biomedical Informatics, Biostatistics, and Medical Epidemiology, University of Missouri (MU) School of Medicine; [6]Department of Medicine, Division of Cardiovascular Medicine, University of Missouri (MU) School of Medicine, Columbia MO, USA.

BACKGROUND

The exposome encompasses the cumulative range of exposures across the life course, including social, psychosocial, behavioral, and environmental factors that may jointly shape health. These multidimensional exposures have been linked to depression and obesity, two conditions contributing substantially to global disease burden.

Exposomic influences may operate through interconnected behavioral and clinical pathways. Sleep disorders and substance use disorders (SUDs) are particularly relevant as potentially modifiable intermediate phenotypes linking broader social, behavioral and psychosocial exposures with subsequent mental and metabolic health.

Rather than acting independently, components of the exposome may form complex dependency structures with intermediate phenotypes and health outcomes. Understanding these interdependencies requires approaches capable of evaluating multiple exposures and clinical conditions simultaneously while accounting for their temporal relationships.

However, the dependency structure connecting the social, behavioral and psychosocial exposome with incident mental and metabolic health remains incompletely characterized. We therefore applied a stability-based causal discovery framework to identify robust dependency structures among multidimensional exposomic factors, sleep disorders, SUDs, depression, and obesity in longitudinal All of Us data.

OBJECTIVE/PURPOSE

Aim 1: Identify stable dependency structures linking the measured social, behavioral, and psychosocial exposome with incident depression and obesity.

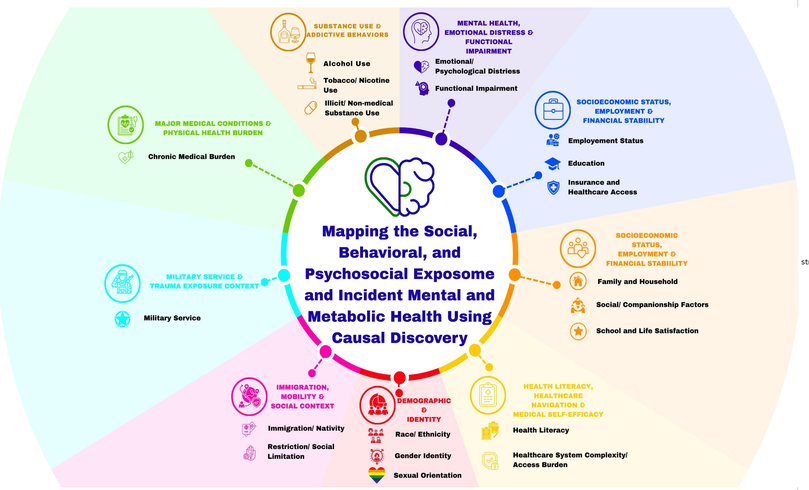
Aim 2: Characterize sleep disorders and substance use disorders (SUDs) as candidate intermediary nodes within exposome–health pathways.

Aim 3: Assess the stability, reproducibility, and directional consistency of exposome–health dependency structures across multiple causal discovery algorithms, bootstrap samples, and analytic specifications.

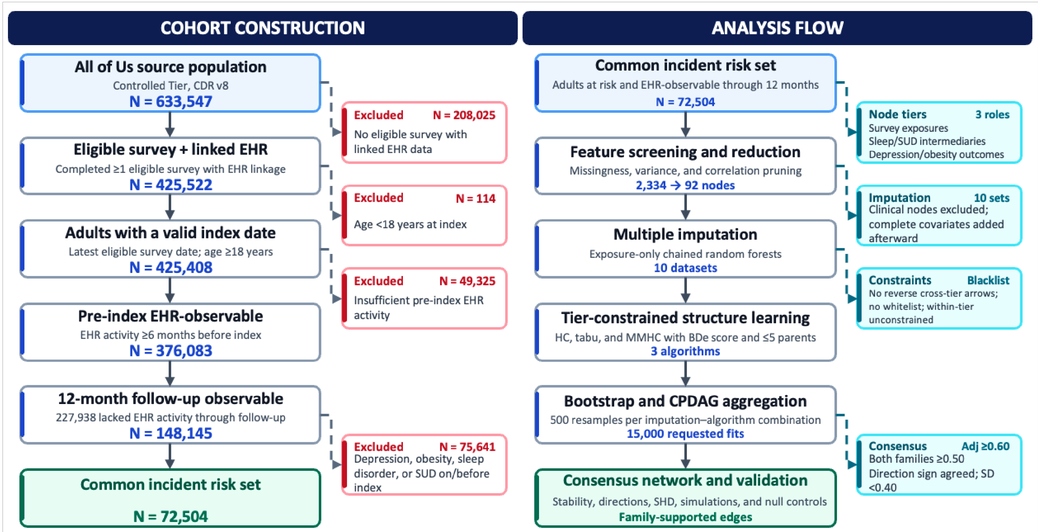
Aim 4: Identify stable and potentially modifiable exposome–health pathways to generate hypotheses for future causal inference, precision prevention, and intervention research.

RESULTS

Figure 1. Summary of Exposure Domains in final DAG



METHODOLOGY



Joint washout categories overlapped: depression, 40,505; obesity, 40,595; sleep disorder, 39,877; SUD, 5,282. Participants without incident events were retained and coded as zero.

HC = hill climbing; MMHC = max–min hill climbing; CPDAG = completed partially directed acyclic graph; SHD = Structural Hamming Distance; SUD = substance use disorder. Directions were classified as design-constrained, data-learned, or unresolved.

RESULTS

Figure 2. Final Consensus DAG of Depression and Obesity Pathways

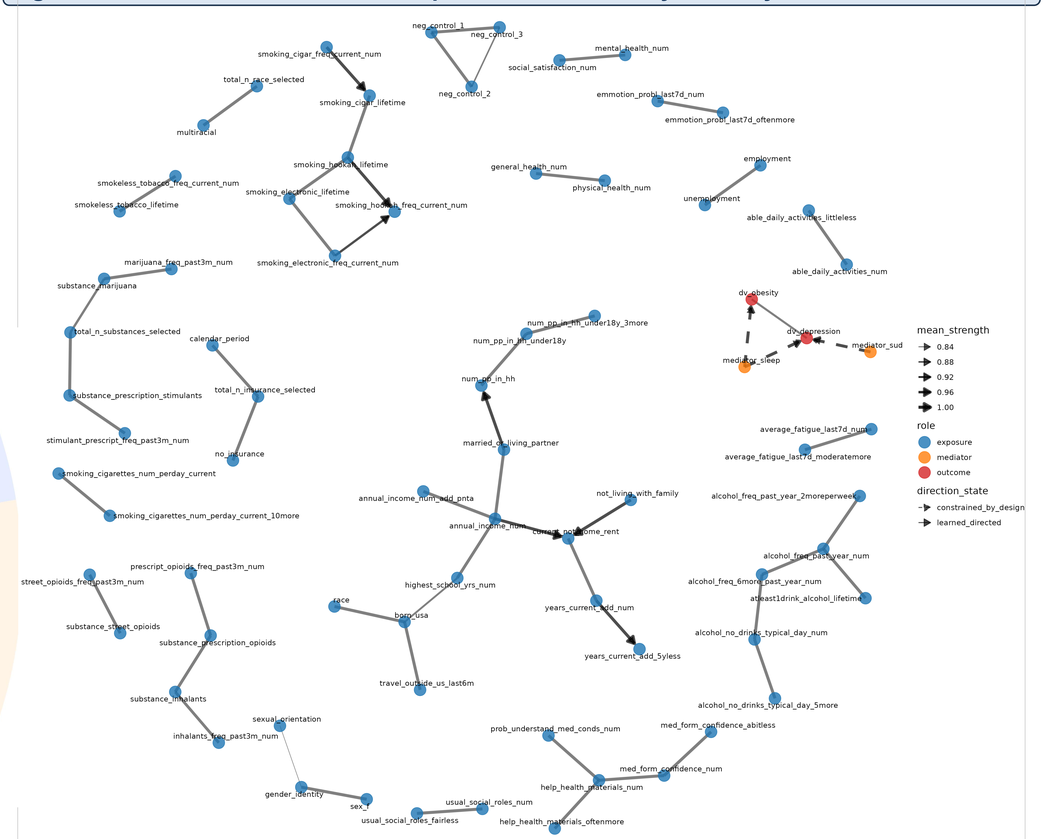


Table 1. Descriptive characteristics of final cohort

Characteristic	Overall (N = 72,504)	Missing (%)
Age at index (years), mean (SD)	50.96 (17.41)	0
Sex assigned at birth [Female], n (%)	42,341 (59.1)	1.2
Non-Hispanic Black, n (%)	17,783 (25.0)	1.9
Non-Hispanic White, n (%)	35,354 (49.7)	1.9
Hispanic, n (%)	13,735 (19.3)	1.9
Incident depression, n (%)	2,041 (2.8)	0
Incident obesity, n (%)	2,273 (3.1)	0
Incident sleep disorder, n (%)	2,116 (2.9)	0
Incident SUD, n (%)	531 (0.7)	0

Table 2. Structural Hamming Distance (SHD)

	HC	TABU	MMHC
HC	0	22	179
TABU	22	0	181
MMHC	179	181	0

- HC and Tabu (score-based methods) generated similar consensus network **structures** (SHD = 22).
- MMHC (hybrid algorithm) showed substantial structural differences compared with HC and Tabu (SHD = 179-181).

CONCLUSIONS

- Stable dependencies were identified primarily among clinical conditions, including sleep disorder–depression, sleep disorder–obesity, SUD–depression, and depression–obesity.
- No measured social, behavioral, or psychosocial exposomic factor showed a stable connection to the clinical nodes under the prespecified consensus criteria.
- Clinical adjacencies should not be interpreted as established causal pathways or mediation: three directions were imposed by temporal background constraints, while depression–obesity remained directionally unresolved.
- The stability-first framework distinguished reproducible adjacency, learned orientation, and design-constrained direction, supporting cautious interpretation of exposome–health dependency structures.

LIMITATIONS

- Selected EHR-observable cohort may limit generalizability.
- First recorded diagnoses may not reflect true disease onset.
- Low event rates may limit stable structure recovery.
- Clinical nodes shared the same follow-up window, limiting temporal and directional inference.
- Survey measures capture only part of the broader exposome.

FUTURE DIRECTIONS

- Expand to broader and longitudinal exposomic measures.
- Establish temporally separated exposure, intermediary, and outcome windows.
- Replicate findings using external cohorts and alternative phenotypes.
- Move from discovery to causal effect estimation and causal machine learning for potentially modifiable pathways.

REFERENCES

1. Visoki E, Moore TM, Zhang X, et al. Classification of suicide attempt risk using environmental and lifestyle factors in 3 large youth cohorts. JAMA Psychiatry. 2024;81(10):1020-1029. doi:10.1001/jamapsychiatry.2024.1887.
2. Luppino FS, de Wit LM, Bouvy PF, Stijnen T, Cuijpers P, Penninx BWJH, et al. Overweight, obesity, and depression: a systematic review and meta-analysis of longitudinal studies. Arch Gen Psychiatry. 2010;67(3):220-229. doi:10.1001/archgenpsychiatry.2010.2.
3. Scutari M. Learning Bayesian networks with the bnlearn R package. J Stat Softw. 2010;35(3):1-22. doi:10.18637/jss.v035.i03.
4. Tsamardinos I, Brown LE, Aliferis CF. The max-min hill-climbing Bayesian network structure learning algorithm. Mach Learn. 2006;65(1):31-78. doi:10.1007/s10994-006-6889-7.

ACKNOWLEDGEMENTS & DISCLAIMER

This research is, in part, funded by the National Institutes of Health (NIH) AIM-AHEAD Program Agreement NO. 1OT2OD032581 and by an NIH Research Project Grant (R01) under Award Number 1R01MH134886-01A1. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the NIH.

We gratefully acknowledge All of Us participants for their contributions, without whom this research would not have been possible. We also thank the National Institutes of Health's All of Us Research Program for making available the participant data examined in this study. The content is solely the responsibility of the authors and does not necessarily represent the official views of the All of Us Research Program or the National Institutes of Health.